The Environment and Development

Chapter 10

Econ 322 – The Global Economy: Trade and Development

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Based on Todaro, P. M. and Smith, C. S. (2020), *Economic Development*, 13th ed., Pearson.

10.1 Environment and Development: The Basic Issues

- **Environmental issues** affect, and are affected by, **economic development**
- Classic **market failures** lead to too much environmental degradation
- Poverty and lack of education may also lead to **non-sustainable use of environmental resources**
- **Global warming** and attendant climate change is a growing concern in developing countries

10.1 The Basic Issues: Topics Covered (1/2)

- **Sustainable development** and **environmental accounting**
- **Population**, resources, and the environment
- **Poverty** and the environment
- **Growth** versus the environment?
- **Rural development** and the environment

10.1 The Basic Issues: Topics Covered (2/2)

- **Urban development** and the environment
- The **global environment** and economy
- Nature and pace of **greenhouse gas-induced climate** change
- **Natural resource-based** livelihoods as a pathway out of poverty: promise and limitations
- The scope of domestic-origin environmental degradation
- Environmental problems have consequences both for health and productivity:
  - Loss of agricultural productivity
  - Unsanitary conditions created by lack of clean water and sanitation
  - Dependence on biomass fuels and pollution; airborne pollutants

The poor as both agents and victims of environmental degradation

- **Victims**:
  - The poor live in **environmentally degraded lands**, which are less expensive because the rich avoid them
  - People living in poverty have **less political clout to reduce pollution** where they live
  - Living in less productive, polluted lands gives the poor **fewer opportunities** to work their way out of poverty
- **Agents**:
  - The **high fertility rate** of people living in poverty
  - **Short time horizon** of the poor (by necessity)
  - Land tenure insecurity
  - Incentives for rainforest resettlement

Environment and Development: Sustainability and Economic Analysis

- **Sustainable development** has been defined as "meeting the needs of the present generation without compromising the well-being of future generations."
- Sustainable development can be studied using **longstanding concepts of economic analysis**. These include the following three tools:
  - Using an **appropriate valuation of future social benefits** – usually valuing the future at a significantly higher rate than does the market
  - Paying proper attention to **market failures** (focusing on externalities and public goods)
  - Explicitly **valuing natural resources as a form of capital stock** rather than just a stream of consumption

Environment and Development: Sustainability

- **Again,** Sustainable development has been defined as "meeting the needs of the present generation without compromising the well-being of future generations."
- **Therefore,** running down the capital stock is not consistent with the idea of sustainability
- Environmental and other forms of capital are **substitutes only to a degree**; eventually, they likely act as complements
- In developing countries, **environmental capital is generally a larger fraction of total capital**
- To know whether environmental capital is increasing or decreasing, we need environmental accounting

Environmental Accounting: Sustainable Net National Income

Sustainable net national income is:

$$NNI^* = GNI - D_m - D_n$$

where

- NNI^* is the **sustainable national income**
- GNI is Gross National Income
- D_m is the depreciation of **manufactured capital** assets
- D_n is the depreciation of **environmental capital**

Running down environmental capital is not sustainable, so its depreciation must be subtracted from income just like the depreciation of machines and buildings.

Sustainable NNI with Restoration and Avertive Expenditure

Expansively, sustainable net national product is:

$$NNI^{**} = GNI - D_m - D_n - R - A$$

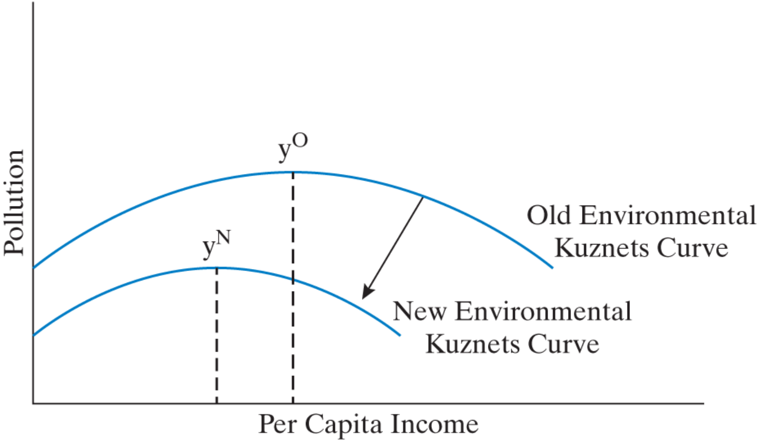
where

- NNI^{**} is the revised (more disaggregated) NNI calculation
- GNI is Gross National Income
- D_m is the depreciation of manufactured capital assets
- D_n is the depreciation of environmental capital
- R is the expenditure needed to **restore** environmental capital
- A is the expenditure required to **avert** the destruction of environmental capital

A Simple Numerical Illustration

- $NNI^{**} = GNI - D_m - D_n - R - A$
- **GNI** = 100 billion rupees
- Depreciation of manufactured capital $D_m = 20$ billion rupees $\Rightarrow$ **NNI** = 80 billion rupees
- Natural resource degradation $D_n = 10$ billion rupees $\Rightarrow$ **NNI*** = 70 billion rupees
- Then unpacking R and A from NNI*:
  - $R = 5$ billion rupees (restoring capital)
  - $A = 5$ billion rupees (averting destruction)
- **NNI**** = 60 billion rupees

Figure 10.1 Hypothetical Income–Pollution Relationship: Environmental Kuznets Curves



Pollution first rises and then falls with per capita income; the curve can shift inward.

Source: Todaro and Smith (2020), Figure 10.1

The Environmental Kuznets Curve: Variations

- Inverted-U observed for such items as lead, sulfur dioxide, particulate matter in the air, and nitrogen oxide
- Decreasing pollution observed throughout, e.g., in water quality and basic sanitation
- Increasing pollution observed throughout – so far and in most cases – in other important items, including greenhouse gases, loss of biodiversity, and landfill volumes
- Note: even if environmental Kuznets curve relationships do hold in the long term, for some items such as loss of biodiversity the damage may be irreversible

Environmental Kuznets Curve: An Aside

- Explanations:
  - Institutional reform (e.g., democratic responsiveness)
  - Public goods provision
  - Private sector action
  - Scope of voluntary action (including NGOs)
- Locus of curves depends on factors such as institutional quality and its evolution over time
- Local/regional vs national/global

Natural Resource Based Livelihoods: Pathways Out of Poverty?

- In low-income countries, **high dependence on natural resources**: agriculture, animal husbandry, fishing, forestry, hunting, and foraging
- But access to the benefits of resources is often very inequitable
- Poor losing control of natural resource commons areas
- Many poor lack farmland, forests, cattle, boats, and equipment
- Common village lands may be "spontaneously" privatized
- Governments may **overlook companies logging, fishing, and mining** without regard to local people or traditional rights
- Governments designate lands "protected," banning livelihoods, while corruption remains; **no incentive to take part in protection**
- An important part of the solution: "pro-poor governance" – empowerment of the poor

Rural Development and the Environment: A Tale of Two Villages

- Representative African village
  - Desertification
  - Low opportunity cost of women's time encourages waste
- Representative South American village
  - Soil erosion
  - Deforestation

Overgrazed, bare land in Rukwa, Tanzania.

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10.2 Global Warming and Climate Change: Scope, Mitigation, and Adaptation

- The Fourth (2007) and Fifth (2014) Benchmark **IPCC Assessment** Reports painted a dire picture for developing economies
- Recent follow up reports have amplified findings and concerns:
  - Summary in the **World Bank 2009 World Development Report**
  - A 2010 U.S. NOAA study found evidence of global warming due to greenhouse gases on all 11 indicators examined
  - 2012 and 2013 "Turn Down the Heat" Reports show severity of consequences for developing countries
  - 2013/2014 Fifth IPCC reports are even stronger than previous: even less uncertainty regarding human causes and severe physical and social consequences of greenhouse gas emissions-based global warming
  - A series of major new reports have been released, and more are scheduled. Monitor http://ipcc.ch/ for ongoing updates
  - The Sixth Assessment Report is now underway, with reports beginning in 2021; see: https://www.ipcc.ch/assessment-report/ar6/

Severity of Climate Change Consequences in SSA and South Asia

- IPCC reports have been a wake-up alarm – though the main response has been to repeatedly hit the snooze button...
- IPCC warns of severe consequences: droughts, desertification, storms, floods, heat waves, deteriorated water resources, reduced crop yields, spread of pests and diseases, sea level rises; loss of grasslands and farmlands
- Sub-Saharan Africa countries will be disproportionately drought-afflicted and suffer the worst impacts on agricultural productivity
- South & Southeast Asia will be disproportionately flood-affected
- Pacific and Indian Ocean bordering states within the cyclone belt will be disproportionately storm-afflicted

Impacts of Climate Change in Developing Countries (IPCC)

- **Adverse health impacts**, including the spread of pests and disease
- Prolonged droughts, expanded desertification
- Increased **severity of storms** with heavy flooding and erosion
- Longer and more severe **heat waves**
- **Reduced summer river flow** and water shortages
- **Agriculture** harmed in tropical and subtropical areas, notably much decreased grain yields
- Lost and **contaminated groundwater**
- Deteriorated freshwater lakes, coastal fisheries, mangroves, and coral reefs
- Coastal flooding
- Loss of essential species such as pollinators and soil organisms
- Forest and crop fires
- Conflicts over natural resources may increase in number and magnitude

The impact of global warming is likely to be hardest on the poorest.

10.2 Global Warming and Climate Change: Scope, Mitigation, and Adaptation

- Historically, **a problem mostly - but not exclusively - caused by developed countries**
- **Growing sources of emissions from developing countries**
  - Rapid industrial growth, especially in Asia: coal plants, autos, etc.
  - Worsened by perverse policies e.g., fossil fuel subsidies
  - Deforestation in developing countries (about 20 %+ of emissions)
- Strategies for mitigation
  - Taxes on carbon
  - Caps on greenhouse gases (generally with "carbon markets")
  - Subsidies to encourage technological progress
- Types of adaptation
  - Planned (or "policy") adaptation
  - Autonomous adaptation

"Planned" and "Autonomous" Adaptation

- **Planned (or policy-driven) adaptation is undertaken by governments**
- **Autonomous adaptation is undertaken by individuals**, families, and communities; examples:
  - Altering crop or livestock varieties
  - Changing livelihoods
  - Increasing exploitation of common-pool resources
  - Moving locally (such as to higher ground)
  - Migrating temporarily or permanently within a country or internationally
- Almost certainly, autonomous adaptation will be the predominant form of adaptation
- Good governance acts as a complement for autonomous adaptation (and, at least, avoids thwarting beneficial responses)

Planned/Autonomous Adaptation Interaction Tradeoffs

- Example 1: Addressing vulnerable low-lying land
  - Policies to vacate low-lying land in densely populated areas subject to inundation increase pressures on adjacent land and increase conflict risks
  - In contrast, in some circumstances, constructing barriers to protect low-lying land might avoid conflict risk
  - But building such barriers may also have unintended environmental, social, and economic consequences
- Example 2: Policies **restricting migration within or across countries** could avoid **tensions and reduce conflict risks**
  - But restrictions could increase risks because migration acts as a key safety valve – one of the most important forms of autonomous adaptation

Adaptation and conflict: Basic examples

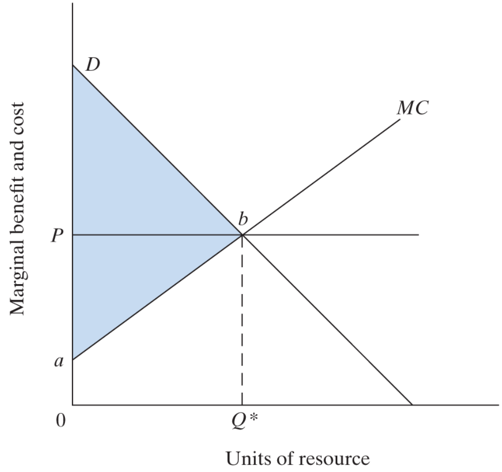
- Effects such as drawing down - or contaminating - neighboring wells
- Out-migration – into already settled areas
- Pastoralists move, encroach on territories of pastoralists or farmers
- Farmers expand territory, encroaching on traditional pastoralist lands
- Communities use neighboring common property, e.g., grazing lands, surface waters

Autonomous adaptation by one household or community often imposes costs on its neighbors – these negative externalities are what turn adaptation into a source of conflict over natural resources.

10.3 Economic Models of Environment Issues

- **Privately owned resources**
- **Inefficiencies** are likely to result from **imperfections in property rights**
- "Perfect" (complete) property rights are characterized by:
  - **Universality** – all resources are privately owned
  - **Exclusivity** or "excludability" – it must be possible to prevent others from benefiting from a privately owned resource
  - **Transferability** – the owner of a resource may sell the resource when desired
  - **Enforceability** – the intended market distribution of the benefits from resources must be enforceable

Figure 10.2 Static Efficiency in Resource Allocation

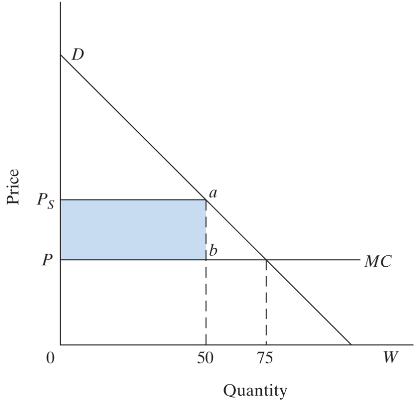


Total net benefit is maximized where marginal cost equals marginal benefit.

Source: Todaro and Smith (2020), Figure 10.2

Figure 10.3 Efficient Resource Allocation Over Time

- Equate the PV of the marginal net benefit of the last unit consumed in each period
- That is, for allocational efficiency the consumer must be indifferent between consuming the last unit in this period or in another period



Allocating a fixed stock between two periods.

Source: Todaro and Smith (2020), Figure 10.3

Worked Example: Two-Period Allocation (1/3)

- **Total available resource**: 100 units (fixed supply, nonrenewable)
- The resource will be allocated over **two periods**: period 1 and period 2
- **Marginal extraction cost**: 0 (to simplify)
- **Inverse demand in each period**:

$$P_t = 100 - Q_t \quad \text{for } t = 1, 2$$

  This implies a linear demand curve, where price decreases as more is consumed.
- **Discount rate**: $r = 0.1$ (10%)

Worked Example: Two-Period Allocation (2/3)

Maximize the **present value (PV)** of total net benefits across both periods:

$$\max_{Q_1, Q_2} \left\{ \int_0^{Q_1} P_1(q)\, dq + \frac{1}{1+r} \int_0^{Q_2} P_2(q)\, dq \right\} \quad \text{subject to} \quad Q_1 + Q_2 = 100$$

Using $P = 100 - Q$, the integral of the demand curve (i.e., consumer surplus) is:

$$\int_0^{Q} (100 - q)\, dq = 100Q - \tfrac{1}{2}Q^2$$

So the objective becomes:

$$\max_{Q_1} \left[100Q_1 - \tfrac{1}{2}Q_1^2 + \frac{1}{1.1}\left(100Q_2 - \tfrac{1}{2}Q_2^2\right) \right]$$

With $Q_2 = 100 - Q_1$, substitute to get:

$$\max_{Q_1} \left[100Q_1 - \tfrac{1}{2}Q_1^2 + \frac{1}{1.1}\left(100(100 - Q_1) - \tfrac{1}{2}(100 - Q_1)^2\right) \right]$$

Worked Example: Two-Period Allocation (3/3)

Differentiate with respect to Q_1:

$$\frac{d}{dQ_1} = 100 - Q_1 - \frac{1}{1.1}\left[-100 + (100 - Q_1)\right] = 100 - Q_1 - \frac{1}{1.1}(Q_1)$$

Set equal to zero:

$$100 - Q_1\left(1 + \frac{1}{1.1}\right) = 0 \;\Rightarrow\; Q_1 = \frac{100}{1 + \frac{1}{1.1}} \approx 52.4 \;\Rightarrow\; Q_2 = 100 - 52.4 = 47.6$$

$Q_1 > Q_2$ because period-2 benefits are discounted at $r = 10\%$; at the optimum the PV of the marginal net benefit is equal in both periods.

10.3 Common Property Resources

- **Inefficiencies** may arise because a resource is **not privately owned**
- Too much labor is used on the **common property** compared to other activities – the so-called **tragedy of the commons**: "too many cows," too many trees cut, too many fish caught; labor input captures each such problem
- Users **fail to take account of an externality**: as each uses more of the common resource, the average return is lowered for other users
- Possibly **too little investment** in the common resource, because others have access to some of the returns

Artisanal fishing pirogues on a Senegalese beach: an open-access fishery.

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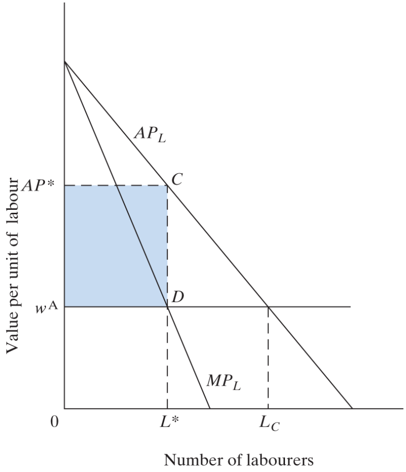
Figure 10.4 Common Property Resources and Misallocation



With open access, labor enters until the average product equals the wage, beyond the efficient level.

Source: Todaro and Smith (2020), Figure 10.4

Tragedy of the Commons: A Simple Model (1/3)

Mathematical setup:

- $i = 1, 2, \ldots, n$: index of herders
- x_i: number of cattle herder i puts on the common pasture
- Total grazing pressure:

$$X = \sum_{i=1}^{n} x_i$$

- Revenue per cow (diminishes as grazing increases):

$$R(X) = a - bX, \quad a, b > 0$$

Tragedy of the Commons: A Simple Model (2/3)

Each herder's private objective:

$$\pi_i = x_i \cdot R(X) = x_i\Big(a - b\sum_{j=1}^{n} x_j\Big)$$

Taking other herders' x_j ($j \neq i$) as fixed, herder i chooses x_i to maximize π_i. Writing $S_{-i} = \sum_{j\neq i} x_j$, the first-order condition is:

$$\frac{d\pi_i}{dx_i} = a - 2bx_i - bS_{-i} = 0 \;\Rightarrow\; x_i^* = \frac{a - bS_{-i}}{2b}$$

Assuming a **symmetric Nash equilibrium** ($x_i = x$ for all i), $S_{-i} = (n-1)x$; solving:

$$x^* = \frac{a}{b(n+1)}, \qquad X^* = nx^* = \frac{an}{b(n+1)}$$

Tragedy of the Commons: A Simple Model (3/3)

A social planner maximizes **total welfare**:

$$W = X(a - bX) \;\Rightarrow\; \frac{dW}{dX} = a - 2bX = 0 \;\Rightarrow\; X^{\text{social}} = \frac{a}{2b}$$

$$X^* = \frac{a}{b} \cdot \frac{n}{n+1} \;>\; \frac{a}{2b} = X^{\text{social}} \quad \text{for } n \geq 2$$

Each herder ignores the fall in revenue per cow that its extra cattle impose on the others, so the commons is overgrazed; the more herders, the closer X^* gets to a/b, where revenue per cow is zero.

The "Tragedy of the Commons": Policy Options

- Problem: Overuse of, and underinvestment in, common property resources
- Direct, external government solution is costly and likely ineffective in many cases. Two alternatives:
- Traditional Economic Analysis Solution: **Privatization**
  - Small farmers can benefit from extended tenancy or ownership
  - Small to medium resource firms
  - Larger firms could generate local employment, but the community may lose benefits of the common resource and not be (fully) compensated
  - Traditional models do not directly address equity and income distribution
  - If privatized: who should buy? (Encouraged/assisted with financing to buy traditionally owned land, which may now be nominally government-owned)
- Alternative Solution: **Local social enforcement mechanisms**
  - Traditional societies have sometimes responded effectively with social enforcement mechanisms based on common property design principles
  - Very difficult to achieve and maintain; increasingly threatened

Elinor Ostrom's Common Property Design Principles

Derived from empirical studies of successful commons:

- Clearly defined boundaries of the resource system
- Proportional equivalence between benefits and costs
- Collective-choice arrangements including those affected
- Monitoring, with those who audit accountable to users
- Graduated sanctions
- Conflict-resolution mechanisms
- Recognition of rights to organize
- Nested enterprises when resources are part of larger systems

Elinor Ostrom, 2009 Nobel laureate in economics.

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Ostrom's Common Property Design Principles: Details

- Clearly defined boundaries
  - The boundaries of the resource system (e.g., irrigation system or fishery) and the individuals or households with rights to harvest resource units are clearly defined
- Proportional equivalence between benefits and costs
  - Rules specifying the amount of resource products that a user is allocated are related to local conditions and to rules requiring labor, materials, and money inputs
- Collective-choice arrangements
  - Many of the individuals affected by the harvesting and protection rules are included in the group who can modify these rules
- Monitoring
  - Monitors, who actively audit biophysical conditions and user behavior, are at least partially accountable to the users or are the users themselves

Ostrom's Common Property Design Principles: Details (Continued)

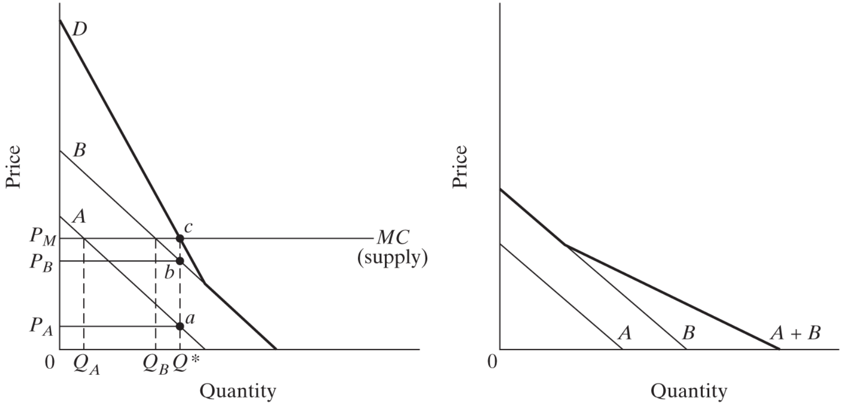
- Graduated sanctions
  - Users who violate rules are likely to receive graduated sanctions (depending on the seriousness and context of the offense) from other users, from officials accountable to these users, or from both
- Conflict resolution mechanisms
  - Users and their officials have rapid access to low-cost, local arenas to resolve conflicts among users or between users and officials
- At least minimal recognition of rights to organize
  - The rights of users to devise their own institutions are not challenged by external governmental authorities, and users have long-term tenure rights to the resource
- For resources that are parts of larger systems: Nested enterprises
  - Appropriation, provision, monitoring, enforcement, conflict resolution, and governance activities are organized in multiple layers of nested enterprises

10.3 Economic Models of Environment Issues (Continued)

- Public goods and bads: regional environmental degradation and the free-rider problem
  - Internalization of externalities is not easy
  - Free rider problems
- Limitations of the public goods framework
  - Pricing mechanism

Regional environmental quality is a public good (and degradation a public bad): each party benefits from others' abatement without paying for it, so everyone free rides and externalities are hard to internalize.

Figure 10.5 Public Goods, Private Goods, and the Free-Rider Problem



(a) Public good (vertical summation) (b) Private good (horizontal summation)

Demand for a public good is summed vertically (a), for a private good horizontally (b).

Source: Todaro and Smith (2020), Figure 10.5

10.4 Urban Development and the Environment

- Environmental problems of urban slums
  - Health-threatening pollutants; unsanitary environmental conditions; serious impact on the poor
- Industrialization and urban air pollution
  - Environmental Kuznets curve; pollution tax; absorptive capacity of the environment
  - Severity of industrial pollution – impact on health
- Problems of congestion, clean water, and sanitation
  - High health and economic costs; a drag on development; impact on the poor
  - Private wells have led to land subsidence and flooding
  - Impact on export earnings

Urban Air Pollution: Delhi

Smog over central Delhi, India – industrialization and urban air pollution.

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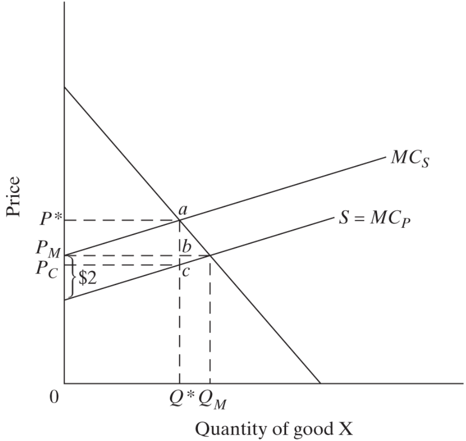
Figure 10.6 Pollution Externalities: Private vs. Social Costs and the Role of Taxation



A pollution tax raises private marginal cost to the social marginal cost, reducing output to the efficient level.

Source: Todaro and Smith (2020), Figure 10.6

Special Features of Adaptation and Resilience Assistance

- *Interactions* between government-led *policy* [or *planned*] *adaptation* and *autonomous adaptation*
- Negative and positive externalities of autonomous adaptation
- Communities must solve *collective action problems*
- Direct and indirect *international political repercussions*
- *Deep uncertainty*
- Role of *real options*
- Adaptation benefits (or costs, with maladaptation) reach across national borders – cross-national *public good (or bad) features*

Special Features of Adaptation and Resilience Assistance (Continued)

- *Natural resource-use responses*
- Endogenous *domestic environmental damage may compound with global-warming* induced (exogenous) climate change
- Needed innovations in financial instruments
- Tensions between building *long-run resilience vs urgent need to adapt* now to current and looming climate impacts now
- Differences between *fostering positive development vs responding to development-in-reverse*

10.5 The Local and Global Costs of Rain Forest Destruction

- Rainforest loss contributes to global warming
- Loss of biodiversity
- **Loss of livelihoods** for people living in poverty who depend upon them
- Much waste is in the process of forest clearing
- Thus, rainforest preservation (and restoration) is a global public good – a restorative mechanism for the environment
- Sustainable management of rain forests is a priority
- Provide funds and debt relief to help enhance biodiversity, and support forest preservation as climate change mitigation

Forest cleared for farmland, Rondônia, Brazil (2020).

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10.6 Policy Options: What Developing Countries Can Do

- Proper resource pricing
- Community involvement
- Clearer property rights and resource ownership
- Improved economic alternatives for the poor
- Improved economic status of women
- Investments that yield returns regardless of the shape of climate change, such as a better road network
- Industrial emissions abatement policies
- Proactive stance toward adapting to climate change

A household solar panel in a village near Mongo, Chad.

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10.6 Policy Options: What Developed Countries Can Do

- What developed countries can do for the global environment
  - Emissions controls, including greenhouse gases
  - Research and development on green technology and pollution control
  - Transfer of technology to developing countries
  - Restrictions on unsustainable production
- Imbalance of global environmental impacts: Earth-from-space photos provide perspective
- How developed countries can help developing countries
  - Lower developing country costs for environmental preservation
  - Trade policies: reduce barriers, subsidies
  - Debt relief and debt-for-nature swaps
  - Development assistance

Chapter summary

- Environmental degradation and development interact; the poor are both its main victims and, out of necessity, agents of it
- Sustainable development means not running down the capital stock: environmental accounting subtracts depreciation of environmental capital from income (NNI*, NNI**)
- Climate change will hit the poorest hardest; mitigation (carbon taxes, caps, subsidies) and planned and autonomous adaptation are both needed
- Imperfect property rights, common property (tragedy of the commons) and public goods (free riding) explain over-use; privatization or Ostrom-style local enforcement are the policy responses
- Policy options range from resource pricing and clearer property rights in developing countries to emissions controls, technology transfer, and debt-for-nature swaps by developed countries

Concepts for Review (1/2)

- Absorptive capacity
- Biodiversity
- Biomass fuels
- Clean technologies
- Climate change
- Common property resource
- Consumer surplus
- Debt-for-nature swap
- Deforestation
- Desertification
- Environmental accounting
- Environmental capital
- Environmental Kuznets curve
- Externality
- Free-rider problem
- Global public good
- Global warming
- Greenhouse gases

Concepts for Review (2/2)

- Internalization
- Marginal cost
- Marginal net benefit
- Pollution tax
- Present value
- Private costs
- Producer surplus
- Property rights
- Public bad
- Public good
- Scarcity rent
- Social cost
- Soil erosion
- Sustainable development
- Sustainable net national income (NNI*)
- Total net benefit